

The background is a dark navy blue. In the top-left corner, there are two overlapping geometric shapes: a blue parallelogram and a light green parallelogram. In the top-right corner, there is a grey, 3D-rendered circuit board pattern. In the bottom-left, there is a circular, semi-transparent inset showing a detailed image of a printed circuit board (PCB) with various electronic components.

Prototype Presentation

The best way to learn a language

TOC

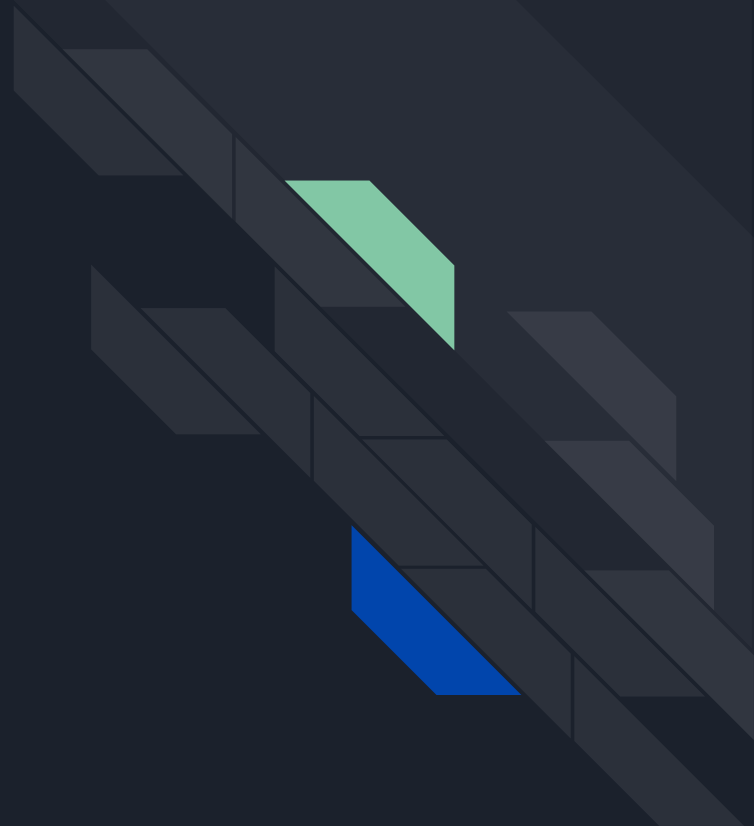
Overview

Understanding the problems

Project objective

Modes

Technologies Used





Overview

Team \$GME is striving to create an innovative way to learn a new language. Our design takes into consideration the different types of learning styles humans have. Our aim is to give everyone a way to except by taking advantage of their learning style but also making sure the user gets multiple versions of repetition in order to move the newly learned vocabulary from short term memory to long term memory. We have used technology that is fairly new but has a large room for growth to ensure that this new method of language learning is engaging enough that an average John Doe can implement in their daily life.



Understanding the problems

- 01 Mobile versions often lack features of the full program
- 02 Language learning apps are easier than their web or software versions
- 03 It's insanely easy to become distracted
- 04 Mobile apps can become a crutch



Project objective

Our application is an Augmented Reality Language Learning app that you can fully immerse yourself into and make it a practical task to learn a new language

Target audience

Our application is rated E for everyone. From a 2 year old to an 80 year old, it is never too early or too late to exercise the mind

Introducing: MoonSpeak

A Modular Language-Learning Application

Review Mode: Put your skills to the test

Learn Mode: Go over the set of words

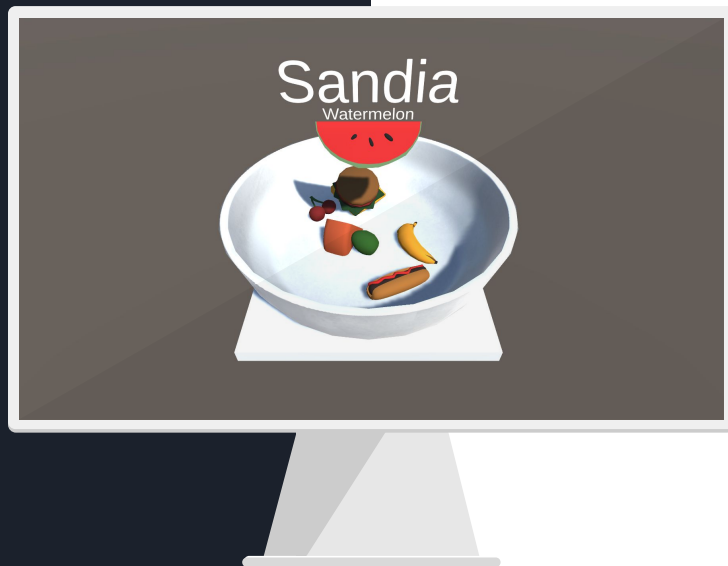
Play: Hardcore mode to test your skills both in speed and retention.

Scores: Keep track of your progress



Learn Mode

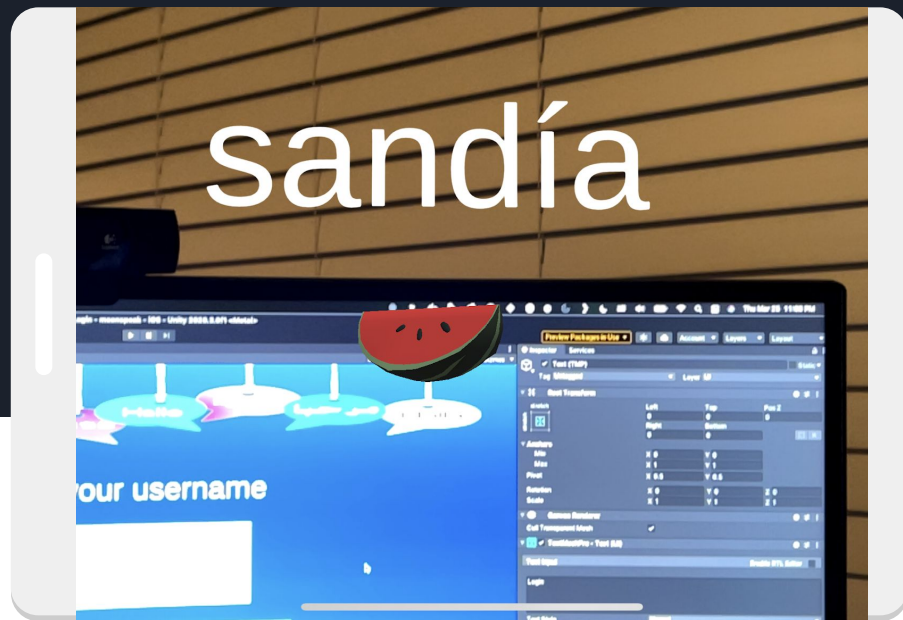
The user can click each object in the fruit bowl to get familiar with the words that they will get tested on in Review Mode



Using the objects that were already imported to the project, we added each one in and have them spawn inside the bowl. If any of the objects falls out, it will respawn after awhile. Right now these are manually added

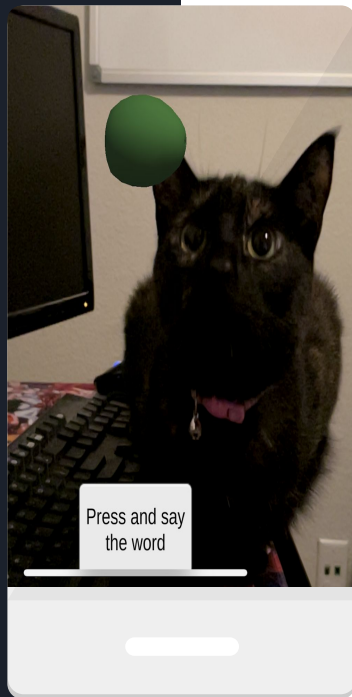
Learn Mode

Our flashcard mode lets you stay at each word for as long as you need. The object will rotate around to get the user familiar with all aspects and when you're ready to move on just swipe



Review Mode

One of our Review modes lets you practice your pronunciations



Through integration with Microsoft Azure's Cognitive Speech Services, our application is able to listen and recognize user speech and match it against the proper pronunciation for the objects contained in our library.

For privacy reasons, users can trigger when to activate and disable the voice recognition feature.

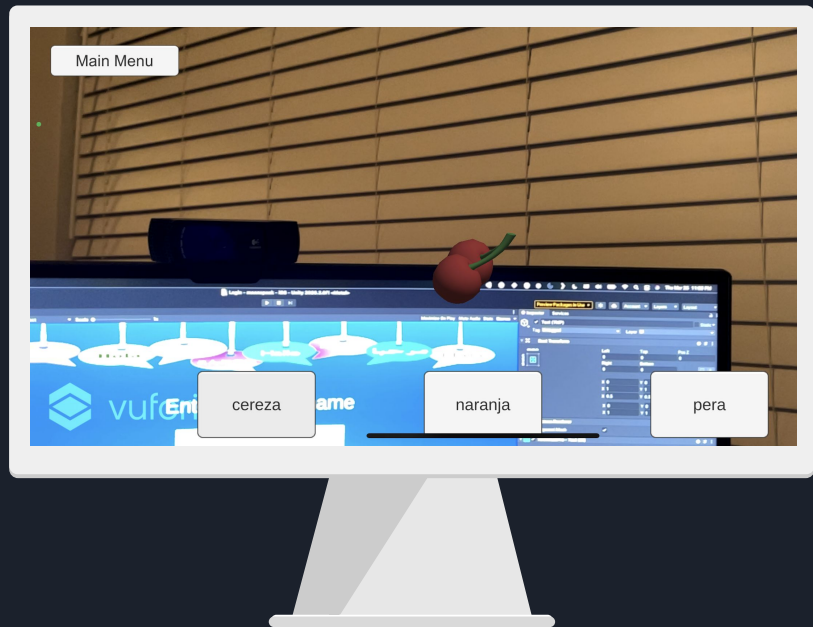
Review Mode



For this mode, the user has to identify the object that matches to the text that appears on the screen

Review Mode

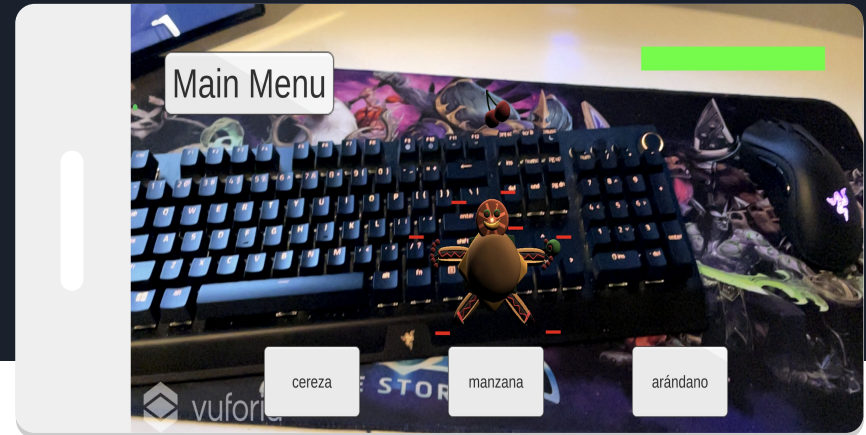
Here the user will receive a single object and has to identify the correct spanish word.



Play Mode

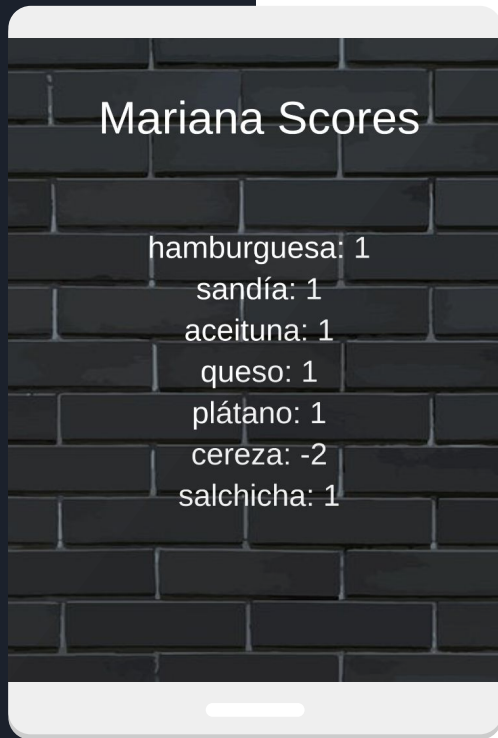
For the users who want a real challenge, they have to quickly match the word with the object that appears above the monsters head and launch an attack to defeat Melvin, before he defeats you

Using all the objects that the user has to learn, we created a Melvin Boss Monster. The rules are stated when the game mode gets chosen. Everything is touch based, the user has to touch the right button on their phone and touch on Melvin's body to do damage



Scores Mode

Keep track of how many times you get the words right or wrong as you progress through your Spanish journey

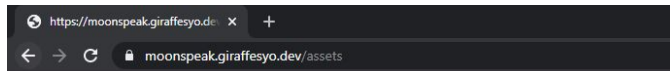


Implemented AWS DynamoDB through a Python Flask API to cache user scores. This just tracks how many times user gets each question correct(+1) and how many times they get it wrong(-1).

DLC Without Updates

Our application dynamically retrieves a compiled asset bundle. We can easily update the content rendered to users through a simple process of building assets in Unity and uploading the files to be used in production.

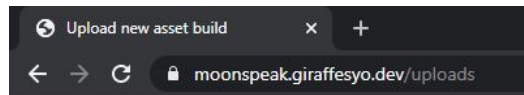
The assets currently in use by distributions for each platform can be previewed



Bucket Contents

StandaloneWindows64/remotevercelprefabs_assets_all_5acc60ce43108bec9adf93c091b3d78b.bundle
iOS/catalog_2021.03.26.03.46.22.hash
iOS/catalog_2021.03.26.03.46.22.json
iOS/remotevercelprefabs_assets_all_7e512e54e3deef21fb22007078c3cb2b.bundle
iOS/remotevercelprefabs_unitybuiltinshaders_e2614c34a0bf5dc61c32e1eb34db8fa4.bundle

And these asset files can be easily updated on a by-platform basis



Upload bundled assets

Build target:

No file chosen



Technologies Used:

Our application leverages cloud services from Amazon Web Services (AWS), Microsoft Azure, and Vercel to facilitate a more fruitful learning experience for end users.

AWS:

- DynamoDB: used for user data persistence, such as scores
- S3: used to host files for dynamic asset loading at runtime

Azure:

- Cognitive Speech Services: used for voice recognition

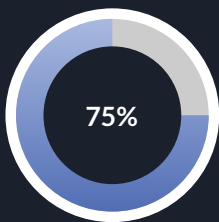
Vercel:

- Project deployment: Python Flask API that interfaces with the aforementioned cloud providers

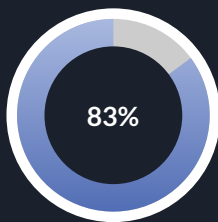
Persona 01

Michael McQuade

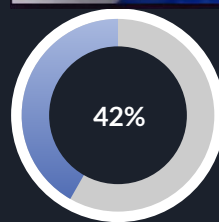
The actual boss monster



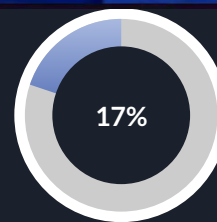
Pizza



Google



Luna



Sleep



Thank you for
Coming to
Our Ted Talk!

